PHYSICS

Higher Level

Set B | Paper 1A and 1B

Challenge mock examination

Time allowed: 120 minutes Maximum mark: 60

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• Paper 1A and Paper 1B share one uninterrupted examination time. Paper 1A has 40 multiple-choice questions; Paper 1B has two compulsory experimental/data questions worth 20 marks altogether.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

• The number in square brackets gives the maximum mark for a written part.

Thriving Scholars challenge practice | Current Physics syllabus | 2026

Paper 1A | Question paper

Paper 1A - Multiple choice

Select one answer, A, B, C or D, for each question. Use the answer grid or record your choices on the portal.

1	2	3	4	5	6	7	8	9	10

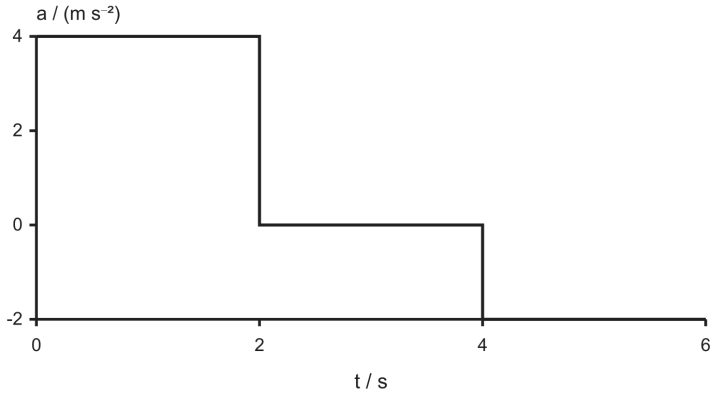
11	12	13	14	15	16	17	18	19	20

21	22	23	24	25	26	27	28	29	30

31	32	33	34	35	36	37	38	39	40

1A Question 1 [1 mark]

A glider has velocity $+2.0\,\text{m}\,\text{s}^{-1}$ at $t = 0$. Its acceleration follows the graph. What is its displacement between $t = 0$ and $t = 6.0\,\text{s}$?



A. $48.0\,\text{m}$

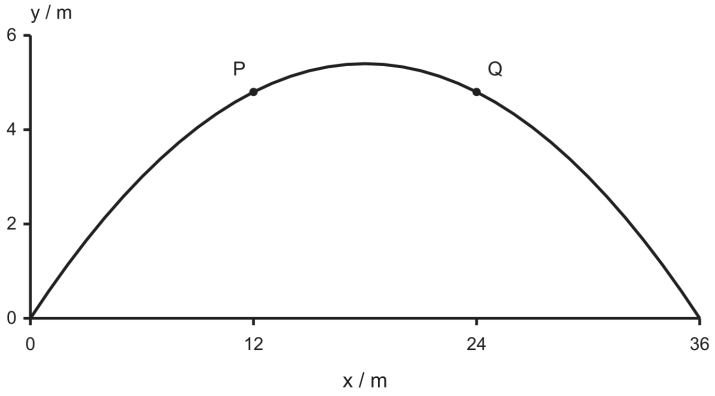
B. $36.0\,\text{m}$

C. $40.0\,\text{m}$

D. $28.0\,\text{m}$

1A Question 2 [1 mark]

A small ball is launched from $(x, y) = (0, 0)$ and moves without air resistance. It passes through $(12.0\,\text{m}, 4.80\,\text{m})$ and $(24.0\,\text{m}, 4.80\,\text{m})$. What is its maximum height above its launch point?



A. $5.40\,\text{m}$

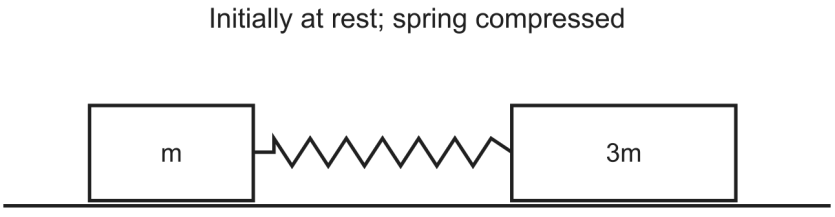
B. $9.60\,\text{m}$

C. $7.20\,\text{m}$

D. $4.80\,\text{m}$

1A Question 3 [1 mark]

Two gliders of masses m and $3m$ are initially at rest on a frictionless track, with a compressed spring between them. They are released. What fraction of the released spring energy becomes kinetic energy of the lighter glider?



A. 3

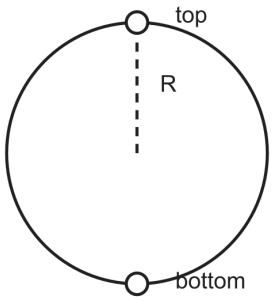
B. $\frac{1}{3}$

C. $\frac{3}{4}$

D. $\frac{1}{4}$

1A Question 4 [1 mark]

A small mass moves on a light string around a complete vertical circle of radius R. Air resistance is negligible and the string remains taut. At the highest point its tension is $2mg$. What is the ratio of the tension at the lowest point to the tension at the highest point?



A. 2

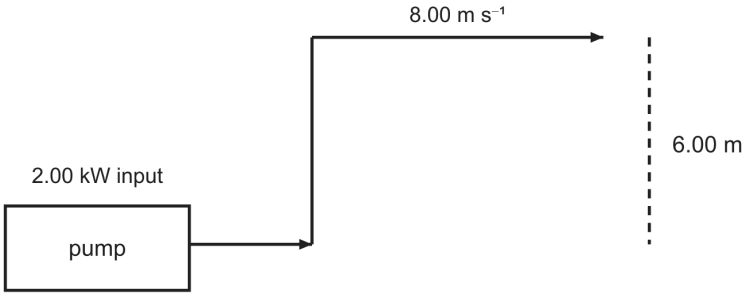
B. 5

C. 3

D. 4

1A Question 5 [1 mark]

A motor uses $2.00\,\text{kW}$ of electrical power and delivers 60.0% of it to water. The water is raised by $6.00\,\text{m}$ and leaves as a jet at $8.00\,\text{m s}^{-1}$. The water entering the pump has negligible speed. Take $g = 9.81\,\text{m s}^{-2}$. What mass of water is delivered each second?



A. $8.16\,\text{kg s}^{-1}$

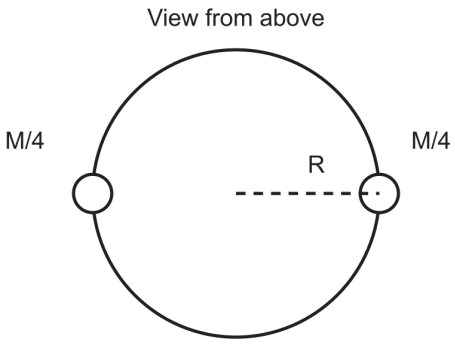
B. $37.5\,\text{kg s}^{-1}$

C. $20.4\,\text{kg s}^{-1}$

D. $13.2\,\text{kg s}^{-1}$

1A Question 6 [1 mark]

A uniform disc of mass M and radius R rotates freely about its central vertical axis. Two small stationary lumps, each of mass $M/4$, fall vertically and stick at opposite points on its rim. The lumps initially have zero angular momentum about the axis. Which row gives the final angular speed and final rotational kinetic energy as fractions of their initial values? Use $I_{\text{disc}} = MR^2/2$.



A. $\frac{1}{2},\ \frac{1}{2}$

B. $\frac{2}{3},\ \frac{2}{3}$

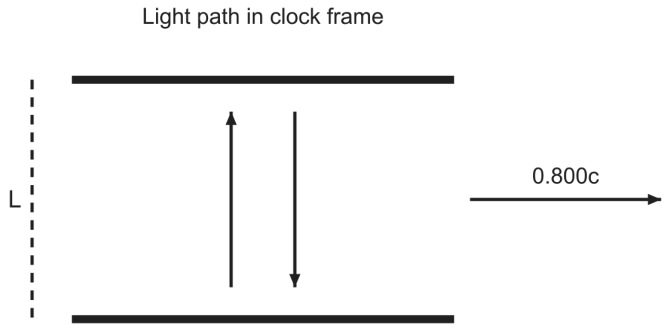
C. $1,\ 1$

D. $\frac{1}{2},\ \frac{1}{4}$

Paper 1A | Question paper

1A Question 7 [1 mark]

A light clock has two parallel mirrors a distance L apart, measured in the clock rest frame. A pulse makes a round trip between them. The clock moves at $0.800c$ parallel to the mirrors relative to Earth. What time does an Earth observer measure for one round trip?



A. $\frac{6L}{5c}$

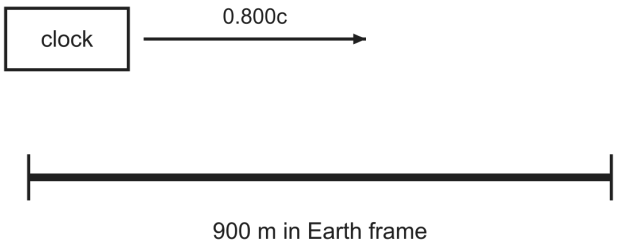
B. $\frac{2L}{c}$

C. $\frac{18L}{5c}$

D. $\frac{10L}{3c}$

1A Question 8 [1 mark]

A platform is $900\,\text{m}$ long in the Earth frame. A spacecraft travelling at $0.800c$ passes both ends. What time elapses on a single clock fixed in the spacecraft between its passage of the first end and the second end? Take $c = 3.00 \times 10^8\,\text{m}\,\text{s}^{-1}$.



A. $3.75\,\mu\text{s}$

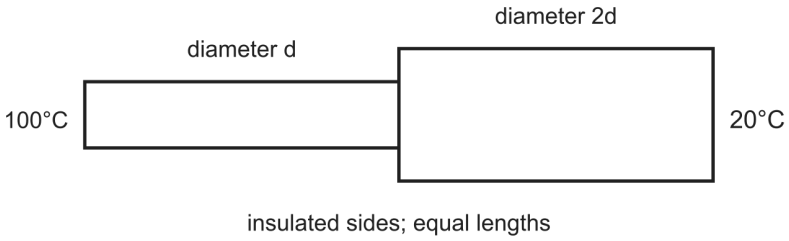
B. $1.80\,\mu\text{s}$

C. $6.25\,\mu\text{s}$

D. $2.25\,\mu\text{s}$

1A Question 9 [1 mark]

Two rods of the same material and length are joined end to end. Their diameters are d and $2d$. Their sides are insulated. The free end of the thin rod is maintained at $100\,^\circ\text{C}$ and the free end of the thick rod at $20\,^\circ\text{C}$. What is the steady interface temperature?



A. $90\,^\circ\text{C}$

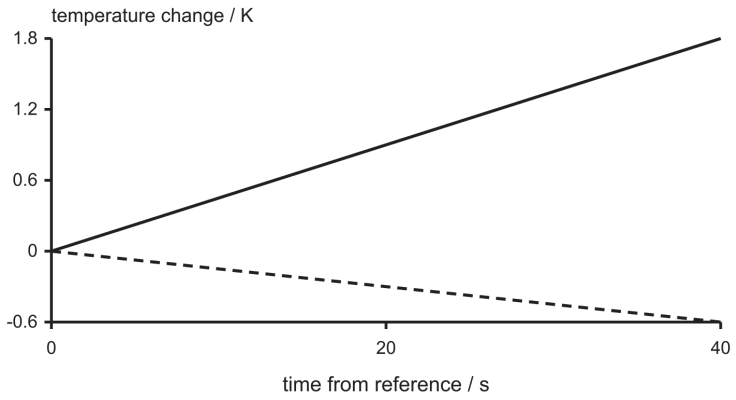
B. $36\,^\circ\text{C}$

C. $60\,^\circ\text{C}$

D. $84\,^\circ\text{C}$

1A Question 10 [1 mark]

A $0.400\,\text{kg}$ liquid sample is in a container of negligible heat capacity. At one temperature it cools at $0.0150\,\text{K}\,\text{s}^{-1}$ with the heater off. At the same temperature, a $100\,\text{W}$ heater makes it warm at $0.0450\,\text{K}\,\text{s}^{-1}$. Heat loss depends only on temperature. What is the liquid specific heat capacity?



A. $4.17 \times 10^3\,\text{J}\,\text{kg}^{-1}\text{K}^{-1}$

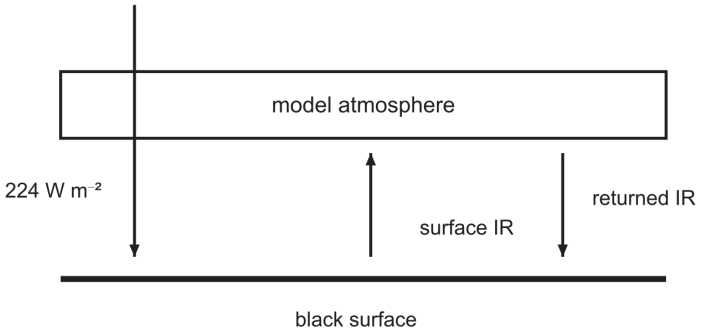
B. $1.67 \times 10^3\,\text{J}\,\text{kg}^{-1}\text{K}^{-1}$

C. $5.56 \times 10^3\,\text{J}\,\text{kg}^{-1}\text{K}^{-1}$

D. $3.33 \times 10^3\,\text{J}\,\text{kg}^{-1}\text{K}^{-1}$

1A Question 11 [1 mark]

A planet surface is black in the infrared and absorbs a global mean solar intensity of $224\,\mathrm{W\,m^{-2}}$. A model atmosphere is transparent to sunlight, absorbs 60.0% of the surface infrared radiation and emits equally upwards and downwards. There is no other energy transfer. What is the equilibrium surface temperature? Use $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}K^{-4}}$.



A. $291\,\mathrm{K}$

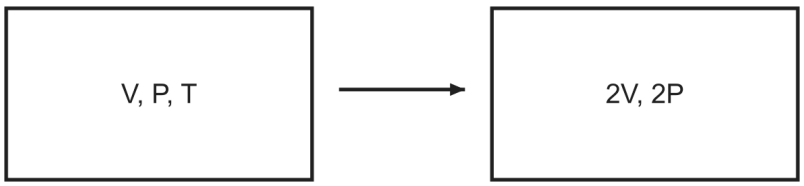
B. $274\,\mathrm{K}$

C. $310\,\mathrm{K}$

D. $252\,\mathrm{K}$

1A Question 12 [1 mark]

Two rigid, insulated vessels contain equal numbers of molecules of the same ideal monatomic gas. Initially, vessel 1 has volume V, pressure P and temperature T. Vessel 2 has volume $2V$ and pressure $2P$. A valve is opened and equilibrium is reached without external work. Which row gives the final pressure and temperature?



A. $\frac{3P}{2}$, $\frac{3T}{2}$

B. $\frac{5P}{2}$, $\frac{5T}{2}$

C. $\frac{5P}{3}$, $2T$

D. $\frac{5P}{3}$, $\frac{5T}{2}$

1A Question 13 [1 mark]

Liquid water is supercooled to $-8.0\,^{\circ}\mathrm{C}$ in an adiabatic vessel. A seed crystal triggers partial freezing, after which ice and water coexist at $0\,^{\circ}\mathrm{C}$. Assume constant liquid specific heat capacity $4200\,\mathrm{J\,kg^{-1}K^{-1}}$ and specific latent heat $3.34\times10^{5}\,\mathrm{J\,kg^{-1}}$. What fraction of the initial water mass freezes?

A. 0.899

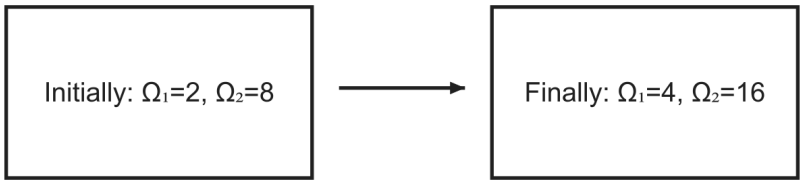
B. 0.101

C. 0.025

D. 1.00

1A Question 14 [1 mark]

Two weakly interacting subsystems have initial multiplicities $\Omega_1 = 2$ and $\Omega_2 = 8$. After an allowed energy redistribution they have $\Omega_1 = 4$ and $\Omega_2 = 16$. Using $S = k_\mathrm{B} \ln \Omega$, what is the change in total entropy?



A. $k_\mathrm{B} \ln 2$

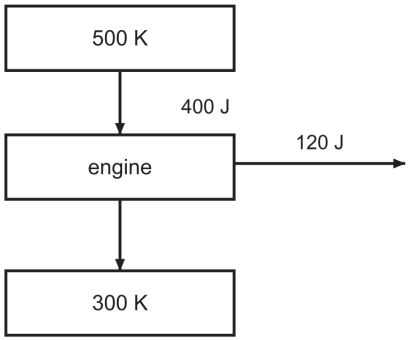
B. $k_\mathrm{B} \ln 32$

C. $k_\mathrm{B} \ln 4$

D. $k_\mathrm{B} \ln 8$

1A Question 15 [1 mark]

In one cycle, an engine takes $400\,\mathrm{J}$ from a reservoir at $500\,\mathrm{K}$ and produces $120\,\mathrm{J}$ of work. Its other reservoir is at $300\,\mathrm{K}$. What is the entropy change of the two reservoirs together?



A. $0.133\,\mathrm{J\,K^{-1}}$

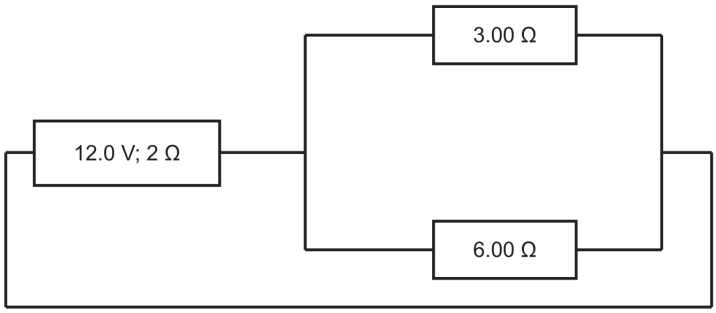
B. 0

C. $0.800\,\mathrm{J\,K^{-1}}$

D. $0.400\,\mathrm{J\,K^{-1}}$

1A Question 16 [1 mark]

A cell has emf $12.0\,\text{V}$ and internal resistance $2.00\,\Omega$. The external load is a $3.00\,\Omega$ resistor in parallel with a $6.00\,\Omega$ resistor. The $3.00\,\Omega$ branch is disconnected. What is the new total external power divided by the original total external power?



A. $\frac{1}{4}$

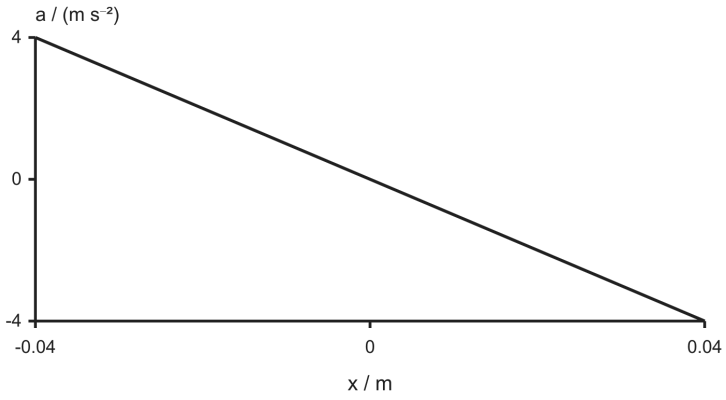
B. $\frac{1}{2}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

1A Question 17 [1 mark]

An oscillator has the acceleration–displacement graph shown. It is released from rest at its positive maximum displacement. When does its speed first reach $\sqrt{3}/2$ of its maximum speed?



A. $0.105\,\text{s}$

B. $0.0524\,\text{s}$

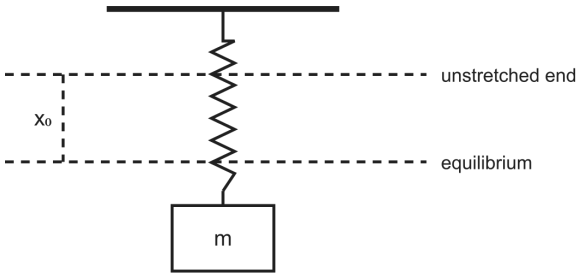
C. $0.157\,\text{s}$

D. $0.314\,\text{s}$

Paper 1A | Question paper

1A Question 18 [1 mark]

A mass hangs from a vertical spring. The equilibrium extension is x_0 and its oscillation amplitude about equilibrium is $0.750x_0$. At an instant when the actual spring extension is $1.50x_0$, what fraction of the total oscillation energy is kinetic energy? The spring remains stretched.



Instant shown: extension 1.50 x_0

A. $\frac{1}{9}$

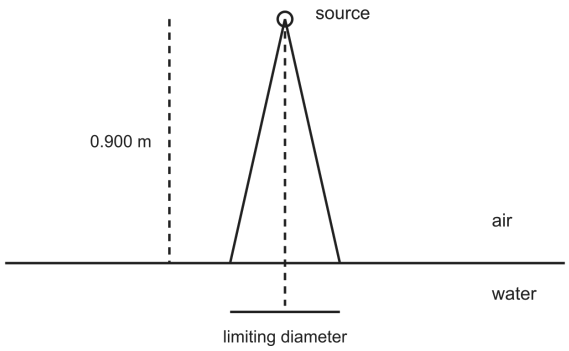
B. $\frac{3}{4}$

C. $\frac{5}{9}$

D. $\frac{4}{9}$

1A Question 19 [1 mark]

A point sound source is $0.900\,\text{m}$ above a flat water surface. Sound speeds are $330\,\text{m}\,\text{s}^{-1}$ in air and $1500\,\text{m}\,\text{s}^{-1}$ in water. In a geometric-acoustics model, only incident rays inside a limiting cone can form propagating transmitted rays. What is the limiting diameter of this cone footprint on the water? Ignore diffraction and treat both media as non-dispersive.



A. $0.396\,\text{m}$

B. $0.406\,\text{m}$

C. $0.203\,\text{m}$

D. $0.198\,\text{m}$

1A Question 20 [1 mark]

A transparent solid has a critical angle of 40.0° when surrounded by air of refractive index 1.00. A ray inside the solid meets a surface at 50.0° to the normal. The solid is immersed in a liquid. What is the limiting liquid refractive index below which total internal reflection still occurs?

A. 1.31

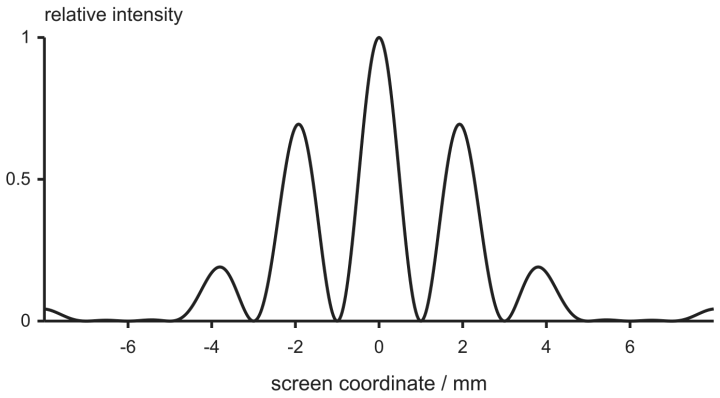
B. 0.839

C. 1.19

D. 1.56

1A Question 21 [1 mark]

A double-slit pattern has adjacent interference maxima separated by $2.00\,\text{mm}$ on a distant screen. The first single-slit diffraction minima are at $\pm 6.00\,\text{mm}$ from the centre. How many bright interference maxima lie strictly inside the central diffraction maximum?



A. 6

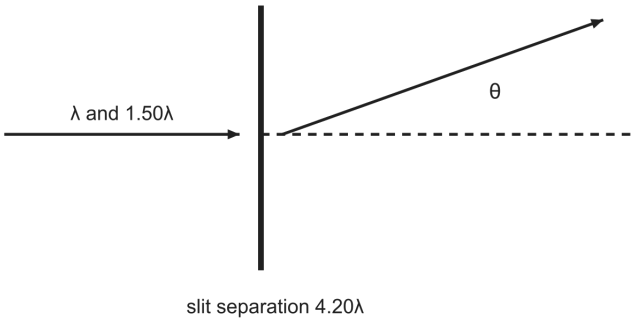
B. 3

C. 7

D. 5

1A Question 22 [1 mark]

Light containing wavelengths λ and 1.50λ is normally incident on a grating of slit separation 4.20λ. At what positive angle is the first non-central maximum common to both wavelengths?



A. 90.0°

B. 20.9°

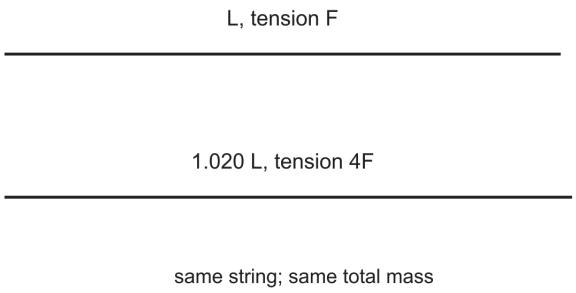
C. 45.6°

D. 30.0°

Paper 1A | Question paper

1A Question 23 [1 mark]

A uniform string is stretched so that its vibrating length increases by 2.00% and its tension becomes four times the original value. Its mass is unchanged and remains uniformly distributed. The wave speed is $v = \sqrt{F/\mu}$, where F is tension and μ is mass per unit length. What is the new fundamental frequency divided by the original fundamental frequency?



A. 1.96

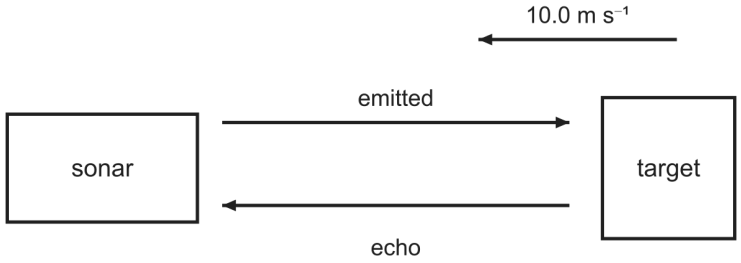
B. 1.98

C. 2.00

D. 2.04

1A Question 24 [1 mark]

A stationary sonar source also receives the reflected wave from a target approaching directly at $10.0\,\mathrm{m\,s^{-1}}$. The source frequency is $40.0\,\mathrm{kHz}$ and wave speed in the medium is $340\,\mathrm{m\,s^{-1}}$. Reflection is coherent and elastic: it preserves frequency in the target rest frame. What beat frequency is formed between the emitted signal and its echo? Here beat frequency means the absolute difference of the two frequencies.



A. $2.42\,\mathrm{kHz}$

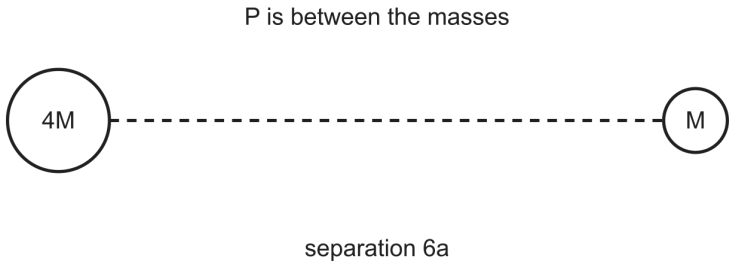
B. $1.18\,\mathrm{kHz}$

C. $1.21\,\mathrm{kHz}$

D. $2.35\,\mathrm{kHz}$

1A Question 25 [1 mark]

Two fixed, isolated point masses $4M$ and M are separated by $6a$. Point P lies between them where their gravitational fields cancel. What minimum speed must a small test particle have at P to escape to infinity? Ignore motion of the source masses.



A. $\sqrt{3GM/(2a)}$

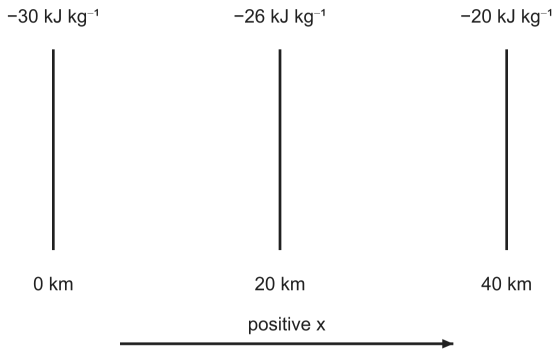
B. $\sqrt{3GM/a}$

C. $\sqrt{GM/a}$

D. 0

1A Question 26 [1 mark]

Along a radial line, the gravitational potentials at $x = 0$, $20\,\text{km}$ and $40\,\text{km}$ are -30, -26 and $-20\,\text{kJ kg}^{-1}$. Use a central-difference estimate to find the gravitational field at $x = 20\,\text{km}$. Positive x is to the right.



A. $0.250\,\text{m s}^{-2}$ right

B. $0.125\,\text{m s}^{-2}$ left

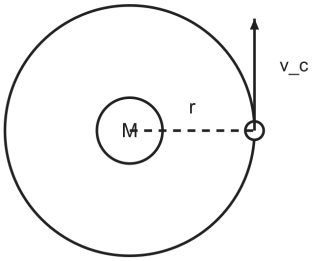
C. $250\,\text{m s}^{-2}$ left

D. $0.250\,\text{m s}^{-2}$ left

Paper 1A | Question paper

1A Question 27 [1 mark]

A satellite moves in a circular orbit at speed v_c around a fixed spherical planet. A brief internal explosion splits it into two equal-mass fragments. One fragment is instantaneously at rest relative to the planet. What is the speed of the other fragment at infinity? Ignore interactions between the fragments, take their masses as negligible relative to the planet and neglect mass loss in the explosion.



Initial circular orbit before explosion

A. $\sqrt{2}\,v_c$

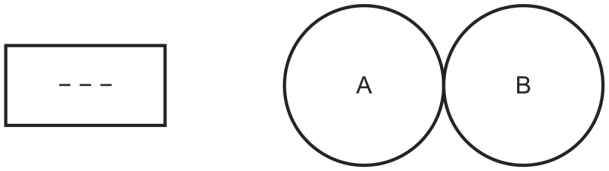
B. 0

C. v_c

D. $2v_c$

1A Question 28 [1 mark]

Two identical neutral conducting spheres A and B touch each other on insulating supports. A negatively charged rod is held near A without touching it. The spheres are separated while the rod remains in place, and then the rod is removed. Which statement is correct?



Spheres initially touch; supports insulate

A. A is negative and B is positive.

B. A is positive and B is negative, with equal charge magnitudes.

C. A and B are both positive.

D. Both spheres return to zero net charge.

1A Question 29 [1 mark]

An electron has kinetic energy $200\,\text{eV}$ at point P. It moves to point Q, whose electric potential is $50.0\,\text{V}$ higher than that of P. Magnetic forces and energy losses are negligible. What is its speed at Q divided by its speed at P? Non-relativistic mechanics applies.

A. 1.12

B. 0.500

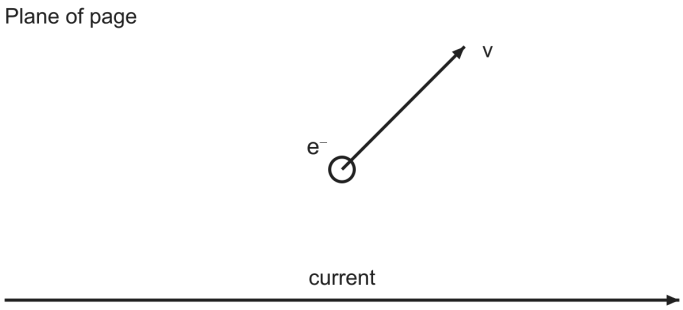
C. 1.25

D. 0.866

Paper 1A | Question paper

1A Question 30 [1 mark]

A long straight wire carries conventional current to the right. An electron above the wire moves diagonally upwards and to the right in the page. What is the direction of the magnetic force on the electron at that instant?



A. Diagonally downwards and to the right.

B. Diagonally downwards and to the left.

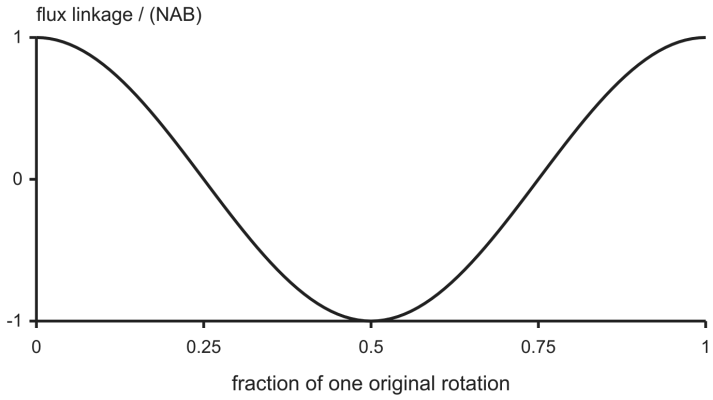
C. Diagonally upwards and to the left.

D. Diagonally upwards and to the right.

Paper 1A | Question paper

1A Question 31 [1 mark]

A coil rotates uniformly in a uniform magnetic field and is connected to a fixed resistor. Its inductance and internal resistance are negligible. Its rotation frequency is doubled while everything else is unchanged. Which row gives the new energy dissipated per rotation and the new mean power, as fractions of their original values?



A. 4, 4

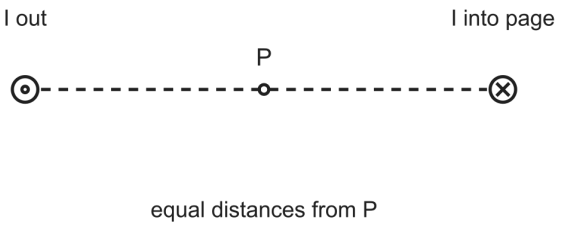
B. 4, 8

C. 2, 4

D. 1, 2

1A Question 32 [1 mark]

Two long parallel wires are perpendicular to the page, equally far from point P midway between them. Initially each carries current I, in opposite directions. One current is doubled. Which row gives the new magnetic field magnitude at P and the new force per unit length between the wires, as fractions of their original values?



A. $\frac{1}{2},\ 2$

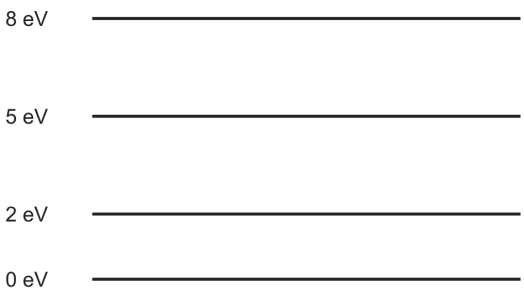
B. $1,\ 2$

C. $2,\ 4$

D. $\frac{3}{2},\ 2$

1A Question 33 [1 mark]

An atom has four energy levels at 0, 2.0, 5.0 and $8.0\,\text{eV}$. Every downward transition is allowed and an excited sample populates all three excited levels. How many distinct emission wavelengths are possible, and how many absorption wavelengths can a separate cold sample in its ground state absorb?



A. $5,\ 6$

B. $5,\ 3$

C. $3,\ 3$

D. $6,\ 3$

1A Question 34 [1 mark]

The energies of a hydrogen-like ion are $E_n = -13.6 Z^2/n^2\,\text{eV}$. A He^+ ion emits a photon in a transition from $n = 4$ to $n = 2$. Which hydrogen transition emits a photon of the same energy?

A. $n = 3 \to n = 1$

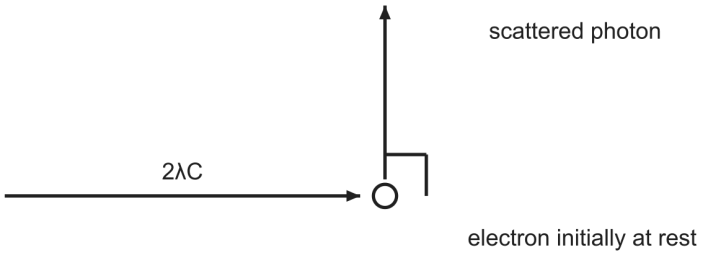
B. $n = 4 \to n = 3$

C. $n = 3 \to n = 2$

D. $n = 2 \to n = 1$

1A Question 35 [1 mark]

A photon of initial wavelength $2\lambda_C$ scatters through 90° from an electron initially at rest, where $\lambda_C = h/(m_e c)$. What fraction of the initial photon energy becomes electron kinetic energy?



A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

1A Question 36 [1 mark]

A metal has stopping potential V under monochromatic light of frequency f. Under light of frequency $2f$, its stopping potential is $3V$. Both frequencies are above threshold. What is the work function of the metal?

A. $\frac{2}{3}hf$

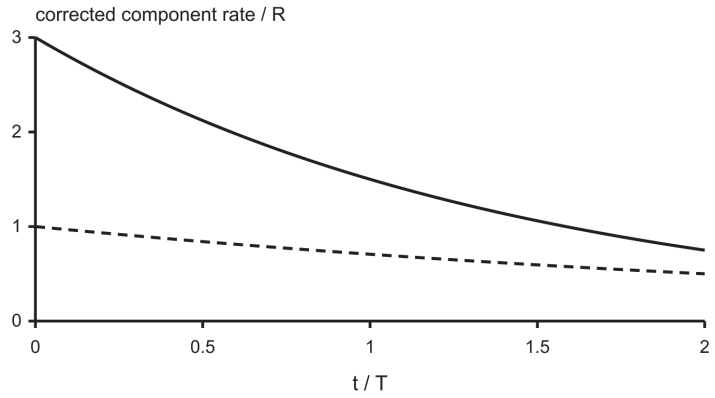
B. $\frac{1}{2}hf$

C. $\frac{1}{4}hf$

D. $\frac{1}{3}hf$

1A Question 37 [1 mark]

A detector measures two independently decaying radionuclides X and Y. Their half-lives are T and $2T$. Initially their background-corrected count-rate contributions are $3R$ and R. Detector efficiencies remain constant. At what time is their combined corrected count rate $2R$?



A. $1.17T$

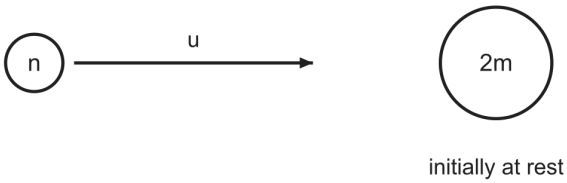
B. $2.00T$

C. $1.50T$

D. $1.00T$

1A Question 38 [1 mark]

A neutron of mass m collides head-on elastically with a stationary deuterium nucleus of mass $2m$. What fraction of the initial neutron kinetic energy is transferred to the nucleus? Treat the particles as free during the collision.



A. $\frac{1}{9}$

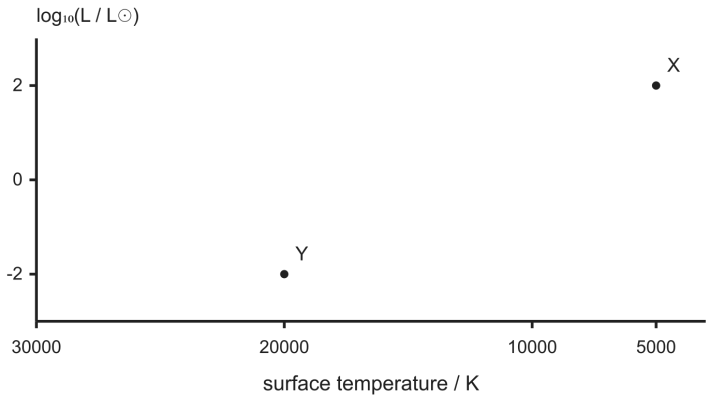
B. $\frac{8}{9}$

C. $\frac{4}{9}$

D. 1

1A Question 39 [1 mark]

Star X has luminosity $100 L_{\odot}$ and surface temperature $5000\,\mathrm{K}$. Star Y has luminosity $0.0100 L_{\odot}$ and surface temperature $20000\,\mathrm{K}$. Both are well modelled as blackbodies. What is R_X / R_Y?



A. 400

B. 25

C. 1600

D. 100

1A Question 40 [1 mark]

Stars A and B have parallaxes 0.0400 and 0.0200 arc-seconds. Their received bolometric fluxes are F and $F/4$ respectively. Star B has twice the surface temperature of A. Assume both stars radiate as blackbodies and extinction is negligible. What is R_B / R_A?

A. $\frac{1}{8}$

B. $\frac{1}{2}$

C. 1

D. $\frac{1}{4}$

1B Question 1 | Question paper

Paper 1B - Experimental and data-based questions

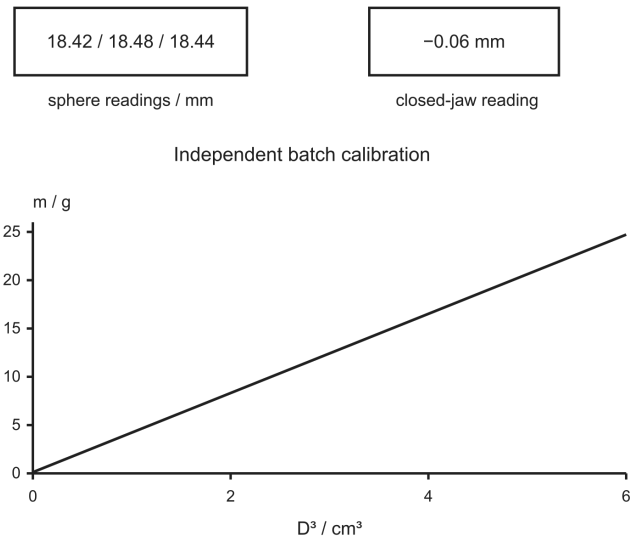
Continue within the same examination time. Answer both questions. Use the data, graphs and uncertainty information provided.

1B Question 1 [10 marks]

A student uses the following laboratory model data to measure the density of a metal sphere. Three caliper readings, in different orientations, are 18.42, 18.48 and $18.44\,\text{mm}$. The closed-jaw reading is $-0.06\,\text{mm}$. The instrument uncertainty of each displayed reading is $\pm 0.02\,\text{mm}$ and the uncertainty in the zero correction is $\pm 0.01\,\text{mm}$.

For this question, estimate the conservative absolute uncertainty in the corrected mean diameter by adding the half-range of the repeated readings, the $0.02\,\text{mm}$ instrument uncertainty and the $0.01\,\text{mm}$ zero-correction uncertainty. The mass is $(26.40 \pm 0.03)\,\text{g}$.

A separate calibrated batch of the same nominal material gives the straight-line fit shown, with $\frac{m}{\text{g}} = 4.10\left(\frac{D}{\text{cm}}\right)^3 + 0.12$.
The mass instrument used for this batch has not had its zero offset removed.



(a) Calculate the corrected mean diameter and its conservative absolute uncertainty. [2]

(b) Use $\rho = 6m/(\pi D^3)$ to calculate the sphere density and its absolute uncertainty. Add fractional uncertainties for products and powers. State an appropriately rounded result in SI units. [3]

(c) Determine the density represented by the gradient of the batch graph. Interpret the nonzero intercept in terms of the batch mass instrument. [2]

(d) The student repeats the same sphere measurements 100 times, using the same unaltered caliper. They claim that all contributions to the quoted diameter uncertainty can therefore be divided by ten. Evaluate the claim and state one measurement change that would address a systematic contribution. [3]

1B Question 2 | Question paper

1B Question 2 [10 marks]

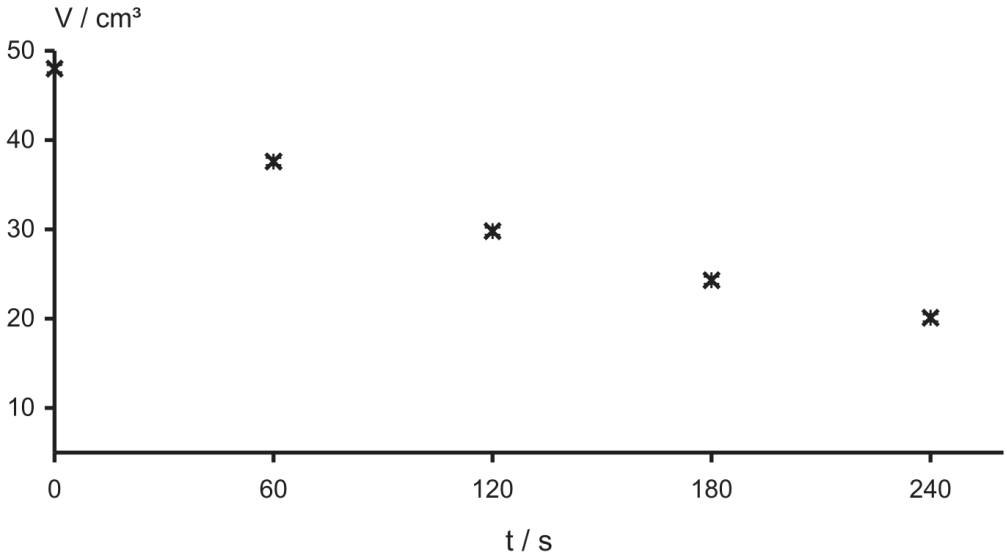
A student investigates a foam layer using the laboratory model data below. Time uncertainty is negligible. Every volume reading has absolute uncertainty $\pm 0.4\,\mathrm{cm}^3$.

t/s	0	60	120	180	240
V/cm^3	48.0	37.6	29.8	24.3	20.1

An independent long-duration measurement estimates a non-collapsing volume $V_\infty = (8.0 \pm 0.3)\,\mathrm{cm}^3$. The proposed model is

$V(t) = V_\infty + U_0 e^{-kt}, \qquad k > 0.$

The student compares the collapsible excess $U = V - V_\infty$ with a radioactive population. The volume is a model analogue, not an actual count of radioactive nuclei.



(a) State a pair of graph axes that would test this model using a straight line. Explain why a graph of $\ln(V/\mathrm{cm}^3)$ against t would not be the appropriate test. [2]

(b) Use the central values at $t = 0$ and $t = 240\,\mathrm{s}$ to determine k and the half-life of the excess volume U. Give the unit of k. [3]

1B Question 2 | Question paper

(c) Using the quoted bounds on the first and last volume readings and on the common value V_∞, determine the largest possible k and hence the smallest possible excess-volume half-life. The same V_∞ must be used at both times. [3]

(d) A later observation at $t = 360\,\text{s}$ is $(18.0 \pm 0.4)\,\text{cm}^3$. Ignoring parameter uncertainty for this diagnostic comparison, calculate the model prediction and state what the discrepancy suggests about extrapolating the model. [2]

End of question paper

PHYSICS

Higher Level

Set B | Paper 1A and 1B

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 1A - Answer key and explanations

1	2	3	4	5	6	7	8	9	10
A	A	C	D	D	A	D	D	B	A

11	12	13	14	15	16	17	18	19	20
B	D	B	C	A	C	A	C	B	C

21	22	23	24	25	26	27	28	29	30
D	C	B	A	B	D	A	B	A	C

31	32	33	34	35	36	37	38	39	40
C	D	B	D	C	B	A	B	C	D

1A Question 1 [1 mark]

Correct answer: A

The velocities at $t = 0, 2, 4, 6\,\text{s}$ are $2, 10, 10, 6\,\text{m s}^{-1}$. Each stage has constant acceleration, so its displacement is mean velocity multiplied by two seconds. The total is $[(2+10)/2](2) + 10(2) + [(10+6)/2](2) = 48.0\,\text{m}$. The area under the acceleration graph gives velocity change, not displacement; omitting a stage or the nonzero starting speed produces the other underestimates.

1A Question 2 [1 mark]

Correct answer: A

The equal-height points are symmetric about $x = 18\,\text{m}$. Writing $y = Kx(36 - x)$ gives $K = 4.80/(12 \times 24) = 1/60\,\text{m}^{-1}$. At the vertex $y_{\max} = 18^2/60 = 5.40\,\text{m}$. The lower value confuses a measured height with the vertex; doubling that height, or assuming the vertex is over one measurement point, fails the two supplied coordinates.

1A Question 3 [1 mark]

Correct answer: C

Zero initial momentum gives $mv = 3mu$, so the lighter glider has speed $3u$. Its energy is $\frac{1}{2}m(3u)^2$, while that of the heavier glider is $\frac{1}{2}(3m)u^2$. Their ratio is $3:1$, so the required fraction is $3/4$. The $1/4$ assigns the larger energy to the larger mass; $1/3$ confuses a ratio with a fraction of the total; 3 is the energy ratio rather than a fraction.

1A Question 4 [1 mark]

Correct answer: D

At the top, $2mg + mg = mv_T^2/R$, giving $v_T^2 = 3gR$. Energy conservation through a drop $2R$ gives $v_B^2 = 7gR$. At the bottom $T_B - mg = mv_B^2/R$, so $T_B = 8mg$ and $T_B/T_T = 4$. Equal-speed assumptions omit the gravitational energy change; omitting the weight in either radial force balance gives the other incorrect ratios.

1A Question 5 [1 mark]

Correct answer: D

The useful power is $1200\,\text{W} = \dot{m}(gh + v^2/2)$. Thus $\dot{m} = 1200/(58.86 + 32.0) = 13.2\,\text{kg}\,\text{s}^{-1}$. Omitting the outgoing kinetic energy gives 20.4; using only kinetic energy gives 37.5; applying efficiency to only one energy transfer is inconsistent.

1A Question 6 [1 mark]

Correct answer: A

The final moment of inertia is MR^2, twice the original. Angular momentum conservation gives $\omega_f = \omega_i/2$. Consequently $K_f/K_i = (I_f/I_i)(\omega_f/\omega_i)^2 = 1/2$. The $1/4$ treats inertia as unchanged; $2/3$ omits one lump; unchanged speed/energy would require an external angular impulse and ignores dissipative capture.

1A Question 7 [1 mark]

Correct answer: D

The proper interval is $\Delta\tau = 2L/c$, since emission and return occur at the same place in the clock frame. With $\gamma = 5/3$, Earth measures $\gamma\Delta\tau = 10L/(3c)$. Dividing by γ reverses time dilation; $2L/c$ is the proper interval; using a linear speed correction is incompatible with the diagonal light path.

1A Question 8 [1 mark]

Correct answer: D

Earth measures $\Delta t = 900/(0.800c) = 3.75\,\mu\text{s}$. Both events occur at the spacecraft clock, so its proper interval is $\Delta\tau = \Delta t/\gamma = 2.25\,\mu\text{s}$. Equivalently the platform contracts to $540\,\text{m}$ in the spacecraft frame. 3.75 is the Earth interval; 6.25 applies time dilation in the wrong direction; 1.80 applies an extra speed/contraction factor.

1A Question 9 [1 mark]

Correct answer: B

Thermal resistance is $L/(kA)$, so the thin rod has four times the resistance of the thick rod. The $80\,\mathrm{K}$ drop divides $4:1$. The thick rod has a $16\,\mathrm{K}$ drop, placing the interface at $36^\circ\mathrm{C}$. 60 assumes equal area; 84 reverses the resistance ratio; 90 treats diameter as a direct temperature ratio.

1A Question 10 [1 mark]

Correct answer: A

Writing $C = mc$, the loss is $0.0150C$. During heating, $100 - 0.0150C = 0.0450C$, so $C = 1667\,\mathrm{J\,K^{-1}}$ and $c = 4167\,\mathrm{J\,kg^{-1}K^{-1}}$. Ignoring loss gives 5556; omitting the sample mass gives 1667 with the wrong physical meaning; subtracting rather than adding the magnitudes of the slopes fails the energy balance.

1A Question 11 [1 mark]

Correct answer: B

If surface emission is $S = \sigma T^4$, the atmosphere returns $0.60S/2 = 0.30S$. Surface balance gives $S = 224 + 0.30S$, hence $S = 320\,\mathrm{W\,m^{-2}}$ and $T = 274\,\mathrm{K}$. Ignoring atmospheric return gives about $251\,\mathrm{K}$; returning all absorbed infrared downwards violates the stipulated equal emission split and overestimates temperature.

1A Question 12 [1 mark]

Correct answer: D

Equal molecule numbers imply $T_2 = 4T$ because $p_2V_2 = 4PV$. Internal energy conservation gives $T_f = (T + 4T)/2 = 2.5T$. Total volume is $3V$ and total molecule number is doubled, so $P_f = 2NkT_f/(3V) = 5P/3$. Arithmetic averaging of pressures is invalid for unequal volumes; the other rows fail either internal-energy conservation or the final gas law.

1A Question 13 [1 mark]

Correct answer: B

Taking liquid water at $0^\circ\mathrm{C}$ as the enthalpy reference, the initial energy per kilogram is $-c(8.0)$ and the final energy per kilogram is $-fL$. Thus $f = c(8.0)/L = 0.101$. Equivalently latent energy released by freezing warms the whole sample. 0.899 is the fraction remaining liquid; complete freezing would release excess energy; 0.025 does not balance the quoted sensible and latent energies.

1A Question 14 [1 mark]

Correct answer: C

Independent multiplicities multiply: $\Omega_i = 2 \times 8 = 16$ and $\Omega_f = 4 \times 16 = 64$. Therefore $\Delta S = k_\mathrm{B} \ln(64/16) = k_\mathrm{B} \ln 4$. Adding multiplicities rather than entropies, using only one subsystem, or confusing a final multiplicity with its ratio to the initial multiplicity gives the other results.

1A Question 15 [1 mark]

Correct answer: A

The engine returns to its initial state and rejects $400 - 120 = 280\,\mathrm{J}$. Thus $\Delta S = -400/500 + 280/300 = 0.133\,\mathrm{J\,K^{-1}}$. Zero would require the reversible work $400(1 - 300/500) = 160\,\mathrm{J}$; the other positive values omit one reservoir or use work as transferred heat.

1A Question 16 [1 mark]

Correct answer: C

Initially $R_L = 2\,\Omega$, $I = 3\,\mathrm{A}$ and $P_L = I^2 R_L = 18\,\mathrm{W}$. Afterwards $R_L = 6\,\Omega$, $I = 1.5\,\mathrm{A}$ and $P_L = 13.5\,\mathrm{W}$. The ratio is $3/4$. Applying the current ratio directly, holding terminal voltage fixed, or counting cell/internal power as load power gives incorrect ratios.

1A Question 17 [1 mark]

Correct answer: A

The slope is $-4/0.040 = -100\,\mathrm{s^{-2}} = -\omega^2$, so $\omega = 10\,\mathrm{rad\,s^{-1}}$. With $x = A\cos\omega t$, $v/v_\mathrm{max} = \sin\omega t$. The first solution of $\sin\omega t = \sqrt{3}/2$ is $t = (\pi/3)/10 = 0.105\,\mathrm{s}$. The other values correspond to different phase fractions, including the quarter- and half-period, rather than the specified first speed.

1A Question 18 [1 mark]

Correct answer: C

The displacement from equilibrium is $y = 1.50x_0 - x_0 = 0.50x_0$, not the whole spring extension. Using the combined gravitational and elastic SHM potential, $K/E = 1 - y^2/A^2 = 1 - (0.50/0.750)^2 = 5/9$. $4/9$ is the potential fraction; the other fractions arise from using absolute spring extension or a linear rather than squared energy relation.

1A Question 19 [1 mark]

Correct answer: B

The critical angle to the normal obeys $\sin\theta_c = v_{\text{air}}/v_{\text{water}} = 0.220$. The limiting radius is $h\tan\theta_c$, so the diameter is $2(0.900)\tan[\sin^{-1}(0.220)] = 0.406\,\text{m}$. 0.203 is the radius; 0.396 substitutes sine for tangent in the footprint geometry; 0.198 combines that small-angle substitution with a radius/diameter error. Rays at the exact limit graze the boundary in water.

1A Question 20 [1 mark]

Correct answer: C

The solid index is $n_s = 1/\sin 40^\circ$. At the limiting condition $n_\ell = n_s \sin 50^\circ = 1.19$; total internal reflection requires a liquid index strictly below this limit. 1.56 is the solid index; 0.839 reverses the ratio; using the old critical angle as unchanged ignores immersion.

1A Question 21 [1 mark]

Correct answer: D

The possible interference maxima are $x = 2m\,\text{mm}$. Inside $-6 < x < 6$, the integers are $m = -2, -1, 0, 1, 2$, giving five. The nominal orders at $\pm 6\,\text{mm}$ coincide with diffraction zeros and are missing. Seven counts those zeros as bright; six omits the central-order counting rule; three counts only one side and the centre.

1A Question 22 [1 mark]

Correct answer: C

Coincidence requires $m\lambda = n(1.50\lambda)$, so the smallest positive integer pair is $m = 3, n = 2$. Hence $\sin\theta = 3\lambda/(4.20\lambda) = 5/7$, giving 45.6°. Using the first order of either colour does not make the positions coincide; 90° is not a solution of either relevant order.

1A Question 23 [1 mark]

Correct answer: B

For fixed total mass m, $\mu = m/L$ and $f = (1/2L)\sqrt{F/\mu} = \tfrac{1}{2}\sqrt{F/(mL)}$. Therefore $f_2/f_1 = \sqrt{4/1.02} = 1.98$. Holding linear density fixed gives about 1.96; ignoring length gives 2.00; multiplying by the length change rather than dividing gives an increase that contradicts the wavelength change.

1A Question 24 [1 mark]

Correct answer: A

The target first acts as a moving observer and then as a moving source: $f_{\text{echo}} = f(v+u)/(v-u)$. The difference is $40.0[(350/330)-1] = 2.42\,\text{kHz}$. A single Doppler shift gives about 1.18 or 1.21; doubling only the observer shift gives 2.35 and misses the second denominator.

1A Question 25 [1 mark]

Correct answer: B

Field cancellation requires $4M/r^2 = M/(6a-r)^2$, so P is $4a$ from $4M$ and $2a$ from M. Its potential is $-4GM/(4a) - GM/(2a) = -3GM/(2a)$. Setting $v^2/2 + \phi = 0$ gives $v = \sqrt{3GM/a}$. Zero force locally does not mean zero potential or zero escape energy; the other roots omit a source or the factor of two from kinetic energy.

1A Question 26 [1 mark]

Correct answer: D

The central gradient is $[(-20)-(-30)] \times 10^3/(40\times10^3) = +0.250\,\text{J kg}^{-1}\text{m}^{-1}$. Field is minus the potential gradient, so it is $0.250\,\text{m s}^{-2}$ left. The rightward answer omits the minus sign; 250 mixes kilometres with metres; 0.125 doubles the baseline without changing the potential difference.

1A Question 27 [1 mark]

Correct answer: A

During the brief explosion, momentum conservation gives $mv_c = (m/2)v$, so the moving fragment initially has speed $2v_c$. Circular motion before the explosion implies $GM/r = v_c^2$. Energy conservation for that fragment after the explosion gives $v_\infty^2 = (2v_c)^2 - 2GM/r = 2v_c^2$, hence $v_\infty = \sqrt{2}v_c$. Zero assumes it is only at escape speed; v_c misses a kinetic-energy factor; $2v_c$ ignores work done against gravity. The explosion can convert internal energy into kinetic energy, so kinetic energy need not be conserved during the split.

1A Question 28 [1 mark]

Correct answer: B

Electrons are repelled away from A into B while the spheres touch. Separation traps an electron deficit on A and an equal excess on B. No charge has entered or left the isolated two-sphere system, so their final net charges are equal and opposite. Both positive would violate charge conservation; reversing signs reverses electron repulsion; removal of the rod cannot transfer electrons across the later gap.

1A Question 29 [1 mark]

Correct answer: A

For an electron, $\Delta U = q\Delta V = -50.0\,\text{eV}$, so $K_Q = 250\,\text{eV}$. Since $v \propto \sqrt{K}$, the speed ratio is $\sqrt{250/200} = 1.12$. 0.866 uses a positive charge sign; 1.25 is the energy ratio, not the speed ratio; 0.500 incorrectly compares the potential change with the initial energy.

1A Question 30 [1 mark]

Correct answer: C

The field above a rightward current is out of the page. For a positive charge, $\mathbf{v} \times \mathbf{B}$ would be down and right. The electron reverses that force, giving up and left, perpendicular to its velocity. Down-right misses the negative charge; the other two directions are not perpendicular to the given velocity or use the wrong wire field.

1A Question 31 [1 mark]

Correct answer: C

Peak emf is $NAB\omega$, so mean resistive power $\varepsilon_0^2/(2R)$ rises by four. A rotation lasts half as long, so energy per rotation rises by two. Equating energy-per-cycle and power ratios ignores the changed duration; treating emf as power gives the first row; the last row applies the frequency factor twice.

1A Question 32 [1 mark]

Correct answer: D

At the midpoint the two opposite-current fields point in the same direction, so their magnitudes add. The field changes from $2B_0$ to $3B_0$, a factor $3/2$. Force per unit length is proportional to $I_1 I_2$, which doubles. Field cancellation would apply to equal currents in the same direction at a midpoint, not the stated opposite directions; neither quantity scales as the square of the changed current alone.

1A Question 33 [1 mark]

Correct answer: B

The six level pairs have energy differences $2, 5, 8, 3, 6, 3\,\text{eV}$: only five are distinct because $3\,\text{eV}$ occurs twice. Ground-state atoms absorb to the three higher levels, giving three wavelengths. Six counts transitions rather than distinct photon energies; three emission lines counts only direct transitions to the ground state; six absorption lines assumes initially excited atoms.

1A Question 34 [1 mark]

Correct answer: D

For He^+, $Z = 2$ and the energy is $13.6(4)(1/4 - 1/16) = 10.2\,\text{eV}$. Hydrogen $2 \to 1$ has $13.6(1 - 1/4) = 10.2\,\text{eV}$. The other hydrogen transitions yield 12.1, 1.89 and $0.661\,\text{eV}$, so matching only the difference in quantum numbers is invalid.

1A Question 35 [1 mark]

Correct answer: C

At $90°$, the Compton shift is λ_C, so the outgoing wavelength is $3\lambda_C$. Its energy is therefore $2/3$ of the incident energy, leaving $1/3$ as electron kinetic energy. $2/3$ is the outgoing photon fraction; $1/2$ treats a fractional wavelength increase as the same energy fraction; $1/6$ is not consistent with energy conservation.

1A Question 36 [1 mark]

Correct answer: B

Einstein equations are $eV = hf - \phi$ and $3eV = 2hf - \phi$. Subtracting gives $2eV = hf$, then $\phi = hf - eV = hf/2$. The other fractions cannot satisfy both equations simultaneously; intensity changes would change emission rate, not these maximum-energy relations.

1A Question 37 [1 mark]

Correct answer: A

The total rate is $3R\,2^{-t/T} + R\,2^{-t/(2T)}$. Let $z = 2^{-t/(2T)}$. The condition becomes $3z^2 + z = 2$, whose physical root is $z = 2/3$. Hence $t/T = -2\ln(2/3)/\ln 2 = 1.17$. The rate halves neither at the shorter nor longer component half-life; $1.50T$ incorrectly averages the half-lives instead of adding the two exponential rates.

1A Question 38 [1 mark]

Correct answer: B

Momentum gives $u = v + 2V$ and relative-speed reversal gives $u = V - v$. Hence $v = -u/3$ and $V = 2u/3$. The nucleus gains $\frac{1}{2}(2m)(2u/3)^2$, which is $8/9$ of $\frac{1}{2}mu^2$. $1/9$ is the remaining neutron-energy fraction; $4/9$ omits the target mass factor; complete transfer requires equal masses in this head-on model.

1A Question 39 [1 mark]

Correct answer: C

From $L = 4\pi R^2 \sigma T^4$, $R_X/R_Y = \sqrt{L_X/L_Y}(T_Y/T_X)^2 = 100(4)^2 = 1600$. The value 100 omits temperature; 400 uses its first power; 25 reverses the temperature factor. The cool luminous star must be vastly larger than the hot faint one.

1A Question 40 [1 mark]

Correct answer: D

B is twice as far away, so $L_B/L_A = (F_B/F_A)(d_B/d_A)^2 = (1/4)(2)^2 = 1$. Equal luminosity and twice the temperature imply $R_B/R_A = (T_A/T_B)^2 = 1/4$. The other choices either confuse flux with luminosity or use a first-power temperature dependence.

Paper 1B - Worked solutions

1B Question 1 [10 marks]

(a) [2 marks]

M1 Mean displayed diameter is $18.4467\,\text{mm}$; subtract the negative zero reading to obtain $D = 18.5067\,\text{mm}$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

A1 Half-range is $0.030\,\text{mm}$, so $\Delta D = 0.030 + 0.020 + 0.010 = 0.060\,\text{mm}$. Report $D = (18.51 \pm 0.06)\,\text{mm}$.

(b) [3 marks]

M1 Convert mass to kilograms and diameter to metres, giving $\rho = 7.95464 \times 10^{3}\,\text{kg}\,\text{m}^{-3}$.

M1 $\Delta\rho/\rho = 0.03/26.40 + 3(0.060/18.5067) = 0.0108626$, so $\Delta\rho = 86.4\,\text{kg}\,\text{m}^{-3}$.

A1 An appropriate report is $(7.95 \pm 0.09) \times 10^{3}\,\text{kg}\,\text{m}^{-3}$, with uncertainty and central value rounded consistently. Accept equivalent justified rounding.

(c) [2 marks]

A1 Since gradient $= \pi\rho/6$, $\rho = (6/\pi)(4.10) = 7.8304\,\text{g}\,\text{cm}^{-3} = 7.83 \times 10^{3}\,\text{kg}\,\text{m}^{-3}$.

R1 The $+0.12\,\text{g}$ intercept indicates a constant positive zero offset of the batch balance in the stated linear model, rather than extra density.

(d) [3 marks]

R1 Repetition can reduce independent random scatter in a mean, but does not justify dividing every uncertainty component by $\sqrt{100}$.

R1 A fixed zero offset/uncertainty and an instrument calibration or resolution limitation do not disappear simply through repeated identical readings.

A1 Use a calibrated reference object to redetermine the zero/calibration, or use an independently calibrated higher-resolution micrometer; state how this addresses the systematic contribution. Rotating the sphere addresses shape/orientation variation but alone does not remove a fixed zero offset.

1B Question 2 [10 marks]

(a) [2 marks]

A1 Plot $\ln[(V - V_\infty)/\mathrm{cm}^3]$ vertically against t/s horizontally. The predicted gradient is $-k$ and intercept is $\ln(U_0/\mathrm{cm}^3)$. Marking guidance: credit a valid subsequent method after an earlier arithmetic or graph-reading error; do not penalize the same numerical error twice. Require a correct physical model and units, and allow consistent use of rounded intermediate values.

R1 The nonzero plateau must first be subtracted. $\ln(V_\infty + U_0 e^{-kt})$ is not a linear function of time for a nonzero plateau.

(b) [3 marks]

M1 $U(0) = 40.0\,\mathrm{cm}^3$ and $U(240) = 12.1\,\mathrm{cm}^3$, so $k = \ln(40.0/12.1)/240$.

A1 $k = 4.98 \times 10^{-3}\,\mathrm{s}^{-1}$.

A1 $t_{1/2} = \ln 2/k = 139\,\mathrm{s}$. This halves U, not total V.

(c) [3 marks]

R1 Maximize the ratio by taking first reading 48.4, last reading 19.7 and the common plateau $8.3\,\mathrm{cm}^3$. Raising the common plateau increases this ratio because the late excess is smaller.

A1 $k_{\max} = \ln[(48.4 - 8.3)/(19.7 - 8.3)]/240 = 5.24 \times 10^{-3}\,\mathrm{s}^{-1}$.

A1 $t_{1/2,\min} = \ln 2/k_{\max} = 132\,\mathrm{s}$. It is invalid to use two different plateau values at the two endpoints.

(d) [2 marks]

A1 $V(360) = 8.0 + 40.0e^{-(0.004981975)(360)} = 14.655\,\mathrm{cm}^3 \approx 14.7\,\mathrm{cm}^3$.

R1 The $3.35\,\mathrm{cm}^3$ central discrepancy is much larger than the stated single-reading uncertainty. In the stipulated diagnostic it challenges this extrapolation; a changing collapse rate, wrong plateau or changing foam structure could be investigated. A short good fit alone does not establish validity for all later times.

End of worked solution book